

When AI Feynman Meets Plasma Turbulence

by Rui Huang

1 Motivation

The typical workflow in natural science research often begins with gathering experimental data and developing mathematical models to describe the underlying mechanisms, that would enable us to make predictions. Over time, as more data is collected using advanced techniques and analyses are refined with more mature mathematics, these models are adjusted to make more accurate predictions. For example, Kepler’s third law was derived by fitting the collected data on Mars’s positions in the sky at different times (Heeffer, 2011). It was this law, not the falling apple from the anecdote, that inspired Newton to formulate his law of gravitation (Hawking, 2002). Later, observed discrepancies in the rate of precession of Mercury’s orbit demanded a better explanation, which Einstein provided with his theory of General Relativity (Einstein, 1916)¹.

Recently, due to the increased amount of data measured by more accurate instruments, this traditional workflow has been forced to undergo a significant transformation (Moseley, 2022). Humans, limited by their cognitive capacity—typically able to handle up to 9 cognitive entities (Miller, 1956; Cowan, 2010) at once—are finding it increasingly challenging to directly interpret the high-dimensional and high-resolution data. This challenge has led to a program synthesis problem known as *symbolic regression* (SR) (Koza, 1994; La Cava et al., 2021; Majumder et al., 2023).

2 What is Symbolic Regression?

In SR, the datasets typically consist of two numerical data matrices: one representing the independent variables $x_1, x_2, \dots, x_n$, and another denoting the dependent variable y . The objective is to discover an accurate symbolic expression for the unknown function f that satisfies $y = f(x_1, x_2, \dots, x_n)$. Mathematically, SR is NP-hard (Virgolin and Pissis, 2022). The computational time required to search for the correct combinations of candidate symbolic terms grows exponentially with the scale of the system when using a naive brute-force searching algorithm. To overcome this, several advanced algorithms have been proposed for SR, including genetic programming (Schmidt and Lipson, 2009), sparse regression (Brunton et al., 2016), Bayesian optimization (Jin et al., 2019), recurrent neural networks (Petersen et al., 2019), and physics-inspired differentiable divide-and-conquer (Wu and Tegmark, 2019; Udrescu and Tegmark, 2020; Udrescu et al., 2020).

From the perspective of computer science, and particularly within the field of machine learning, the development of deep neural networks has long been criticized for its opacity. Although there have been attempts to develop post-hoc methods for explaining black-box model predictions (Lundberg and Lee, 2017), there is a rising priority on the explicit construction of transparent models (Rudin, 2019). In this context, SR not only helps explain

¹Ultimately, science advances by slaying its heroes rather than doubtlessly worshipping them.

black-box neural networks, but also provides interpretable, user-friendly, and controllable surrogate mathematical models that may expose deeper insights, given that mathematics is often considered as the language of the universe (Strogatz, 2019; Tegmark, 2008). From the perspective of science, SR serves as a systematic tool for discovering mathematical rules in data and has been applied in astrophysics (Cranmer et al., 2020), chemistry (Hernandez et al., 2019), and biology (Schmidt and Lipson, 2011), holding great potential to revolutionize various aspects of scientific discovery, especially in fields that rely heavily on phenomenological theories.

3 The Goal of this Course Project

Our primary objective, in the long term, is to develop an advanced turbulent heating model capable of predicting the ion-to-electron heating rate Q_i/Q_e based on fundamental plasma and turbulence parameters (Howes, 2024). We already have a turbulent heating model, Howes (2010) (hereafter referred to as the H10 model). The H10 model is numerically computed from a gyrokinetic turbulence cascade model (Howes et al., 2008) and empirically formulated to provide Q_i/Q_e as a function of the ion plasma beta β_i and the ion-to-electron temperature ratio T_i/T_e . The empirical formula, manually fitted by the human scientist, Gregory Howes, was crafted through a trial-and-error iterative process. Looking ahead, as the complexity of our heating model increases, we may encounter challenges in fitting higher-dimensional data. To address this, one potential improvement could be to systematize the data-fitting process.

The goal of this course project is to have a first look at the performance of some symbolic regression algorithms and the feasibility of extending their use for handling more complicated data in the future. Specifically, this project focuses on applying the physics-inspired divide-and-conquer symbolic regression algorithm, AI Feynman, to the H10 model data. This course project seeks to answer: **Given the H10 dataset, can AI Feynman uncover a more accurate symbolic expression than the human-fitted empirical formula? If yes, could this lead to new physics insights in plasma turbulence heating? If no, why?**

4 Why AI Feynman?

The idea of AI Feynman originates from the work of Wu and Tegmark (2019), which proposes a physics-inspired divide-and-conquer symbolic regression method. This method utilizes the fact that functions in physics and various scientific fields often exhibit some simplifying properties to decompose a harder problem into smaller, easier-to-solve ones with fewer variables. Subsequently, Udrescu and Tegmark (2020) enhances this initial work and applies neural networks (NNs) to uncover the hidden simplicities in datasets. In particular, they selected 100 equations from the Feynman Lectures on Physics, used these equations to generate data, and fed the data into the algorithm. The algorithm managed to recover all of them. Hence, they name their algorithm “AI Feynman”.

As depicted in Fig. 1, their AI Feynman algorithm is structured into several modules. Initially, it applies the dimensional analysis to simplify the problem. If the problem persists,

the algorithm attempts a polynomial fit, acknowledging that many physical functions can be expressed in terms of low-degree polynomials. Should the issue remain unresolved, the algorithm models the target function using a parse tree of limited node types, resorting to a brute-force search. In cases where the problem is still unsolved, AI Feynman algorithm trains a feed-forward, fully-connected NN to obtain an interpolating function for the given datasets. Given that functions in physics tend to be continuous, such an interpolating function usually exists and is approachable. This function allows the exploration of specific symmetries or separabilities, potentially reducing the number of variables or dividing the original dataset into smaller, independently solvable subsets. Furthermore, the algorithm may equalize two input variables to simplify the problem or explore additional transformations, such as higher-order polynomials, exponentials, and trigonometric functions. This iterative process continues until a solution is found. **The AI Feynman workflow closely mirrors the approach a human scientist might take to fit data, combining the physics understanding of a researcher with the computational power of modern machinery.** Therefore, this algorithm could be beneficial in our research on developing the turbulence heating model, where the independent variables are the grid points of plasma and turbulence parameter space, and the heating rate acts as the dependent variable.

Note that the code used in this project is actually AI Feynman 2.0, available at <https://github.com/SJ001/AI-Feynman>, which is based on the paper (Udrescu et al., 2020). Compared to the original AI Feynman, this updated version offers several improvements: (1) it can explore the generalized symmetries; (2) it is more stable to noise; (3) the brute-force search speed is faster; (4) it is capable of identifying not only functions but also distributions; and (5) it outputs Pareto-optimal formulas, which balance accuracy and complexity optimally.

All of the above enhancements, along with the original AI Feynman workflow, hold significant potential for improving performance in this project if properly modified. However, due to time constraints, I have only reviewed the optimization process so far and have not made any modifications to the AI Feynman code. Therefore, in this project, AI Feynman is utilized as a “black box.”

Both the accuracy and the complexity are quantified using the description length, which, loosely speaking, is the number of bits required to describe an error dataset and a symbolic expression and related to Kolmogorov complexity (Kolmogorov, 1963). The specific definition used in the AI Feynman code is given in Wu and Tegmark (2019), which appears to be a simplified version of the standard definition in Grünwald et al. (2005). Fig. 2, which is Figure 1 of Udrescu et al. (2020), displays a Pareto-frontier of four kinetic energy formulas discovered by AI Feynman. Each formula has the highest accuracy for its level of complexity. Here, the labeled “Inaccuracy” in Fig. 2 is also called the “mean error-description-length loss (MEDL)”.

5 The H10 Dataset and the Empirical Formula

As previously mentioned, the H10 dataset is generated from a numerical calculation of a cascade model and consists of three columns: the first column represents the ion plasma beta β_i , with values ranging logarithmically from 0.01 to 100; the second column records the

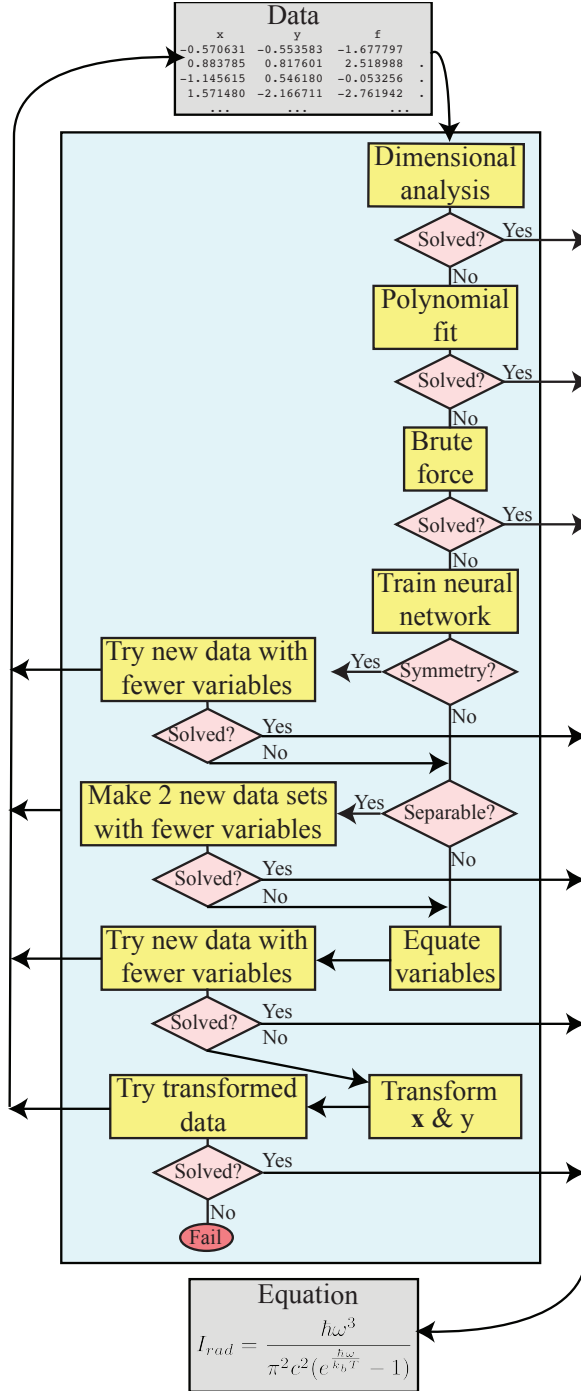


Figure 1: Fig.1 of (Udrescu and Tegmark, 2020), showing the workflow of *AI Feynman* algorithm.

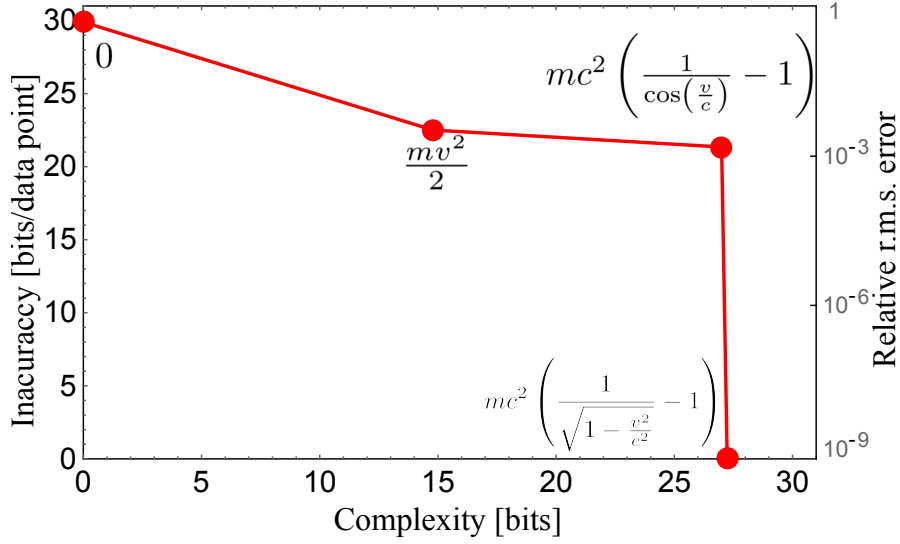


Figure 2: Fig.1 of Udrescu et al. (2020), the Pareto-frontier of the AI Feynman-discovered kinetic energy formulas.

temperature ratio T_i/T_e , which varies logarithmically from 0.2 to 100; and the third column provides the numerical values for the ion-to-electron heating ratio Q_i/Q_e corresponding to each specific β_i and T_i/T_e . The dataset includes 161 samples for β_i and 109 samples for T_i/T_e , resulting in a relatively dense grid. A detailed description of the H10 dataset can be found in Table 1.

The objective is to find a symbolic expression for Q_i/Q_e as a function of β_i and T_i/T_e . It is worthwhile to first examine the features of the data. Fig. 3(a) displays a contour plot of $\log_{10}(Q_i/Q_e)$ across the parameter plane $(\beta_i, T_i/T_e)$ (see also Figure 1 of (Howes, 2010)). This plot reveals that the heating rate Q_i/Q_e predominantly varies with β_i and shows minimal dependence on T_i/T_e . This behavior is succinctly described by the e^{-1/β_i} term² in the fitted empirical formula, i.e. eq.(3) of (Howes, 2010).

$$\frac{Q_i}{Q_e} \left(\beta_i, \frac{T_i}{T_e} \right) = c_1 \frac{c_2^2 + \beta_i^{2-0.2\log_{10}(\frac{T_i}{T_e})}}{c_3^2 + \beta_i^{2-0.2\log_{10}(\frac{T_i}{T_e})}} \sqrt{\frac{m_i T_i}{m_e T_e}} e^{-1/\beta_i}, \quad (1)$$

where $c_1 = 0.92$, $c_2 = 1.6/(T_i/T_e)$, $c_3 = 18 + 5 \log_{10}(T_i/T_e)$.

To visualize the high accuracy of Eq.(1), I present the same contour plot in Fig. 3 (b) as Fig. 3 (a) but is calculated using Eq.(1). In addition, I plot $\log_{10}(Q_i/Q_e)$ against $\log_{10}(\beta_i)$ at $T_i/T_e = 0.2, 1, 100$, using both the original data and the empirical formula, in Fig. 3 (c). The resemblance between Fig. 3 (a) and (b), along with the closely aligned curves in Fig. 3 (c), convincingly illustrates the empirical formula's ability to capture the H10 dataset trends.

²According to the human scientist, Gregory Howes, who conducted this fitting, the term $\sqrt{m_i T_i / (m_e T_e)} e^{-1/\beta_i}$ reveals physics insights and has been derived from the previous studies (Quataert, 1998). Most of his fitting efforts involved varying the values of c_1, c_2, c_3 . This information motivated me to test the AI Feynman fitting by passing $\beta_i, T_i/T_e, \sqrt{m_e T_e / (m_i T_i)} e^{1/\beta_i} Q_i/Q_e$ as the input data. The results are plotted in Fig. 7.

	β_i	T_i/T_e	Q_i/Q_e
count	17551	17551	17551
mean	11.104968	16.375302	1.847498e+01
std	21.164030	23.994394	4.853178e+01
min	0.010000	0.200000	6.688329e-47
25%	0.100000	0.945740	1.936305e-06
50%	1.000000	4.472100	2.052830e-01
75%	10.000000	21.147000	1.060999e+01
max	100.000000	100.000000	3.691333e+02

Table 1: Details of the H10 dataset.

Moreover, I have calculated the complexity and the MEDL for the empirical formula Eq.(1). The complexity is 95.77, and the MEDL is 22.28.

6 Symbolic Expressions from AI Feynman Fitting

This section presents all the symbolic expressions obtained from AI Feynman fitting by plotting $\log_{10}(Q_i/Q_e)$ versus $\log_{10} \beta_i$ at a fixed T_i/T_e value, comparing the fitted expressions with the original data. Since AI Feynman code is treated as “black box” here, the only effort made to improve performance is the data pre-processing. Two types of the data pre-processing were tested:

1. **Input Data Transformation:** Instead of using raw values of $\beta_i, T_i/T_e, Q_i/Q_e$, various transformed forms of the data were passed. For example, the logarithm of the data $\log_{10} \beta_i, \log_{10} T_i/T_e, \log_{10} Q_i/Q_e$, and the dependent variable as Q_i/Q_e divided by the dominate term e^{-1/β_i} rather than simply Q_i/Q_e .
2. **Input Data Segmentation:** As shown in the contour plot in Fig. 3 (a), the system exhibits distinct behaviors between $T_i/T_e > 1$ and $T_i/T_e < 1$, as well as between $Q_i/Q_e > 1$ and $Q_i/Q_e < 1$. Consequently, several tests were conducted using data within specified ranges.

Due to time constraints, I was unable to draw strong conclusions from the plots at least for now. Therefore, I have included all the plots I currently have in this report, for the convenience of potential future discussions with more experienced professionals. A table of contents detailing each data preprocessing trial and the corresponding figure index displaying the results can be found in Table 2.

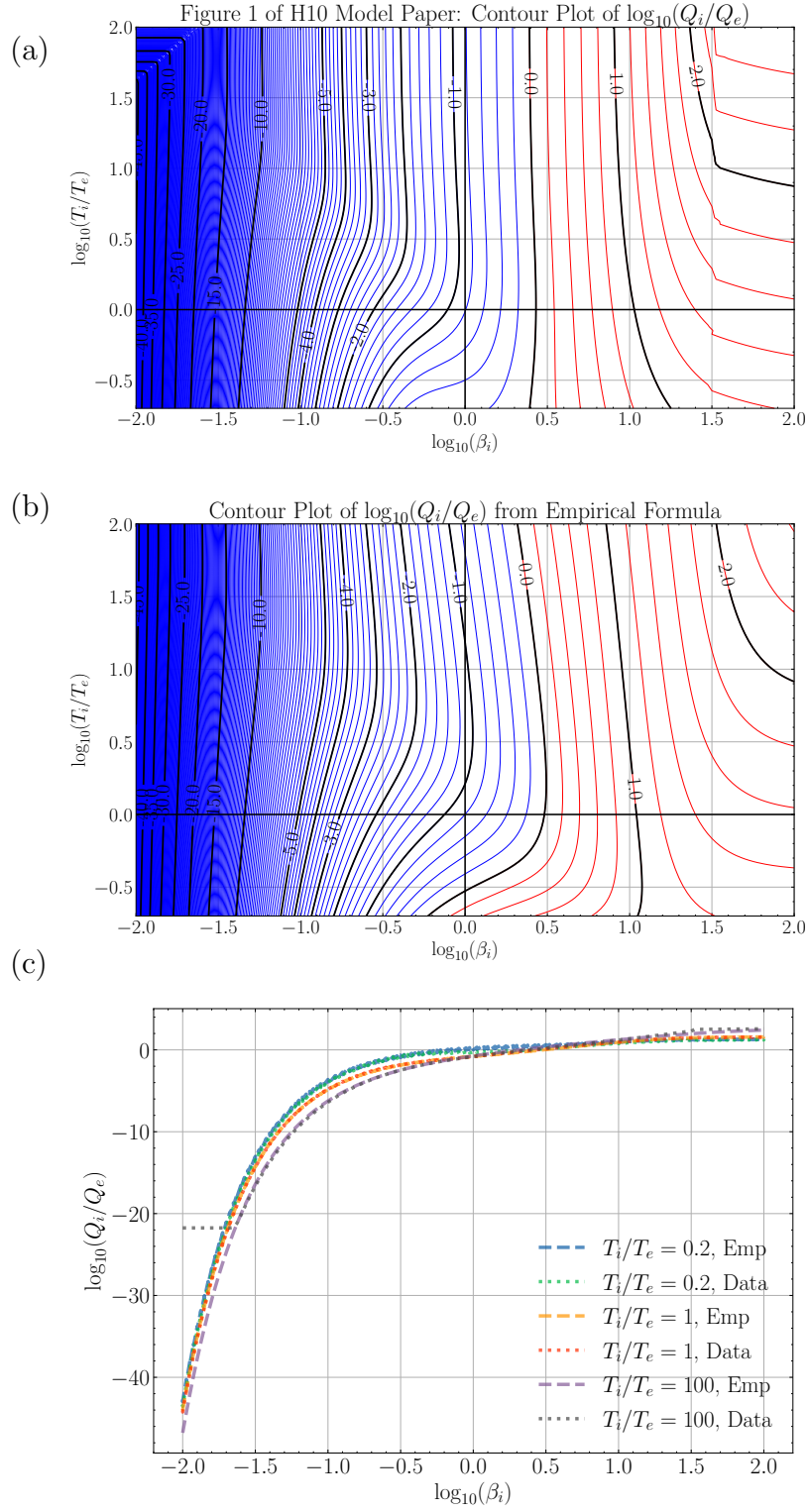


Figure 3: Comparison of the original H10 dataset and the fitted empirical formula Eq.(1).

Figure	Pre-Processing	Input Data	Input Data Range	Plotted Expressions	T_i/T_e
Fig. 4 (a)	None	$\beta_i, T_i/T_e, Q_i/Q_e$	All	All	1
Fig. 4 (b)	None	$\beta_i, T_i/T_e, Q_i/Q_e$	All	Selected	0.2
Fig. 4 (c)	None	$\beta_i, T_i/T_e, Q_i/Q_e$	All	Selected	1
Fig. 4 (d)	None	$\beta_i, T_i/T_e, Q_i/Q_e$	All	Selected	100
Fig. 5 (a)	T	$\log_{10}(\beta_i), \log_{10}(T_i/T_e), \log_{10}(Q_i/Q_e)$	All	All	1
Fig. 5 (b)	T	$\beta_i, \log_{10}(T_i/T_e), Q_i/Q_e$	All	All	1
Fig. 5 (c)	T	$e^{-1/\beta_i}, \log_{10}(T_i/T_e), Q_i/Q_e$	All	All	1
Fig. 5 (d)	T	$\beta_i, T_i/T_e, e^{1/\beta_i}(Q_i/Q_e)$	All	All	1
Fig. 6 (a)	T	$\beta_i, T_i/T_e, e^{1/\beta_i}(Q_i/Q_e)$	All	Selected	0.2
Fig. 6 (b)	T	$\beta_i, T_i/T_e, e^{1/\beta_i}(Q_i/Q_e)$	All	Selected	1
Fig. 6 (c)	T	$\beta_i, T_i/T_e, e^{1/\beta_i}(Q_i/Q_e)$	All	Selected	50
Fig. 6 (d)	T	$\beta_i, T_i/T_e, e^{1/\beta_i}(Q_i/Q_e)$	All	Selected	100
Fig. 7 (a)	T	$\beta_i, T_i/T_e, e^{1/\beta_i} \sqrt{m_e T_e / (m_i T_i)} (Q_i/Q_e)$	All	All	0.2
Fig. 7 (b)	T	$\beta_i, T_i/T_e, e^{1/\beta_i} \sqrt{m_e T_e / (m_i T_i)} (Q_i/Q_e)$	All	All	1
Fig. 7 (c)	T	$\beta_i, T_i/T_e, e^{1/\beta_i} \sqrt{m_e T_e / (m_i T_i)} (Q_i/Q_e)$	All	All	10
Fig. 7 (d)	T	$\beta_i, T_i/T_e, e^{1/\beta_i} \sqrt{m_e T_e / (m_i T_i)} (Q_i/Q_e)$	All	All	100
Fig. 8 (a)	S	$\beta_i, T_i/T_e, Q_i/Q_e$	$T_i/T_e < 1$	All	0.2
Fig. 8 (b)	S	$\beta_i, T_i/T_e, Q_i/Q_e$	$T_i/T_e > 1$	All	10
Fig. 8 (c)	TS	$\beta_i, T_i/T_e, e^{1/\beta_i}(Q_i/Q_e)$	$T_i/T_e < 1$	All	0.2
Fig. 8 (d)	TS	$\beta_i, T_i/T_e, e^{1/\beta_i}(Q_i/Q_e)$	$T_i/T_e > 1$	All	10
Fig. 9 (a)	TS	$\beta_i, T_i/T_e, e^{1/\beta_i}(Q_i/Q_e)$	$Q_i/Q_e < 1$	All	1
Fig. 9 (b)	S	$\beta_i, T_i/T_e, Q_i/Q_e$	$Q_i/Q_e > 1$	All	1
Fig. 10 (a)	S	$\beta_i, T_i/T_e, Q_i/Q_e$	$0.01 < Q_i/Q_e < 100$	All	0.2
Fig. 10 (b)	TS	$\beta_i, T_i/T_e, e^{1/\beta_i}(Q_i/Q_e)$	$0.01 < Q_i/Q_e < 100$	All	0.2
Fig. 10 (c)	S	$\beta_i, T_i/T_e, Q_i/Q_e$	$0.01 < Q_i/Q_e < 100$	All	1
Fig. 10 (d)	TS	$\beta_i, T_i/T_e, e^{1/\beta_i}(Q_i/Q_e)$	$0.01 < Q_i/Q_e < 100$	All	1
Fig. 10 (e)	S	$\beta_i, T_i/T_e, Q_i/Q_e$	$0.01 < Q_i/Q_e < 100$	All	100
Fig. 10 (f)	TS	$\beta_i, T_i/T_e, e^{1/\beta_i}(Q_i/Q_e)$	$0.01 < Q_i/Q_e < 100$	All	100

Table 2: Table of Content of Each Plot of AI Feynman Results. T: Input Data Transformation; S: Input Data Segmentation.

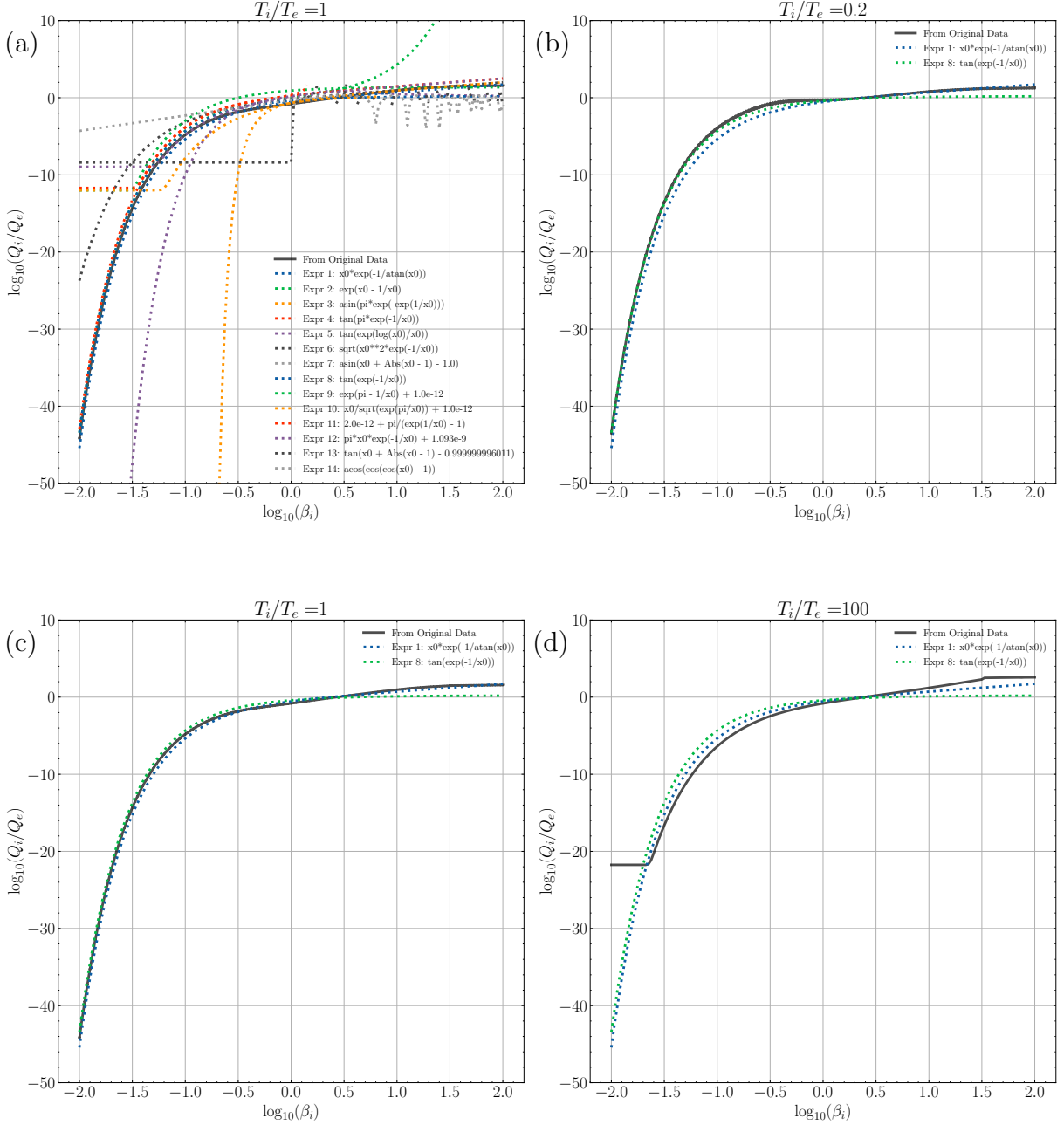


Figure 4: Results from AI Feynman without data pre-processing. Each panel displays $\log_{10}(Q_i/Q_e)$ versus $\log_{10} \beta_i$ at fixed T_i/T_e values, as labeled in the title of each panel. The plots include both the original data and the fitted expressions, detailed in the figure legends. Here $x_0 = \beta_i$, and $x_1 = T_i/T_e$; the form of the expressions follows the syntax of `SymPy`. Panel (a) presents all AI Feynman-generated symbolic expressions at $T_i/T_e = 1$. Panel (b), (c), and (d) show the two best expressions at $T_i/T_e = 0.2, 1, 100$, respectively.

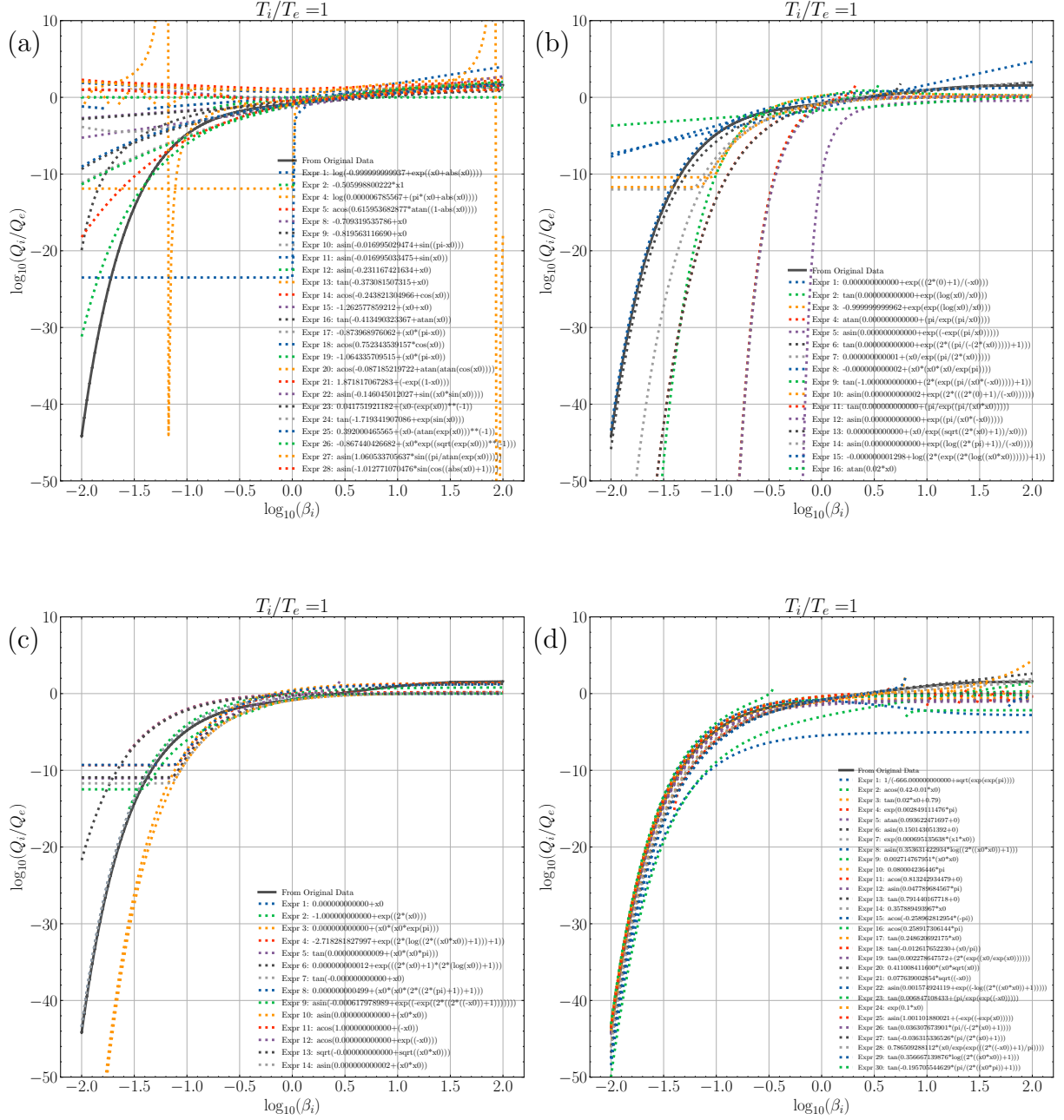


Figure 5: Results from AI Feynman fitting with input data transformation. Each panel follows the same layout as Fig. 4(a). Here, the fitted symbolic expressions are plotted from tests with transformed input data (a): $\log_{10}(\beta_i), \log_{10}(T_i/T_e), \log_{10}(Q_i/Q_e)$; (b): $\beta_i, \log_{10}(T_i/T_e), Q_i/Q_e$; (c): $e^{-1/\beta_i}, \log_{10}(T_i/T_e), Q_i/Q_e$; (d): $\beta_i, T_i/T_e, e^{1/\beta_i}(Q_i/Q_e)$.

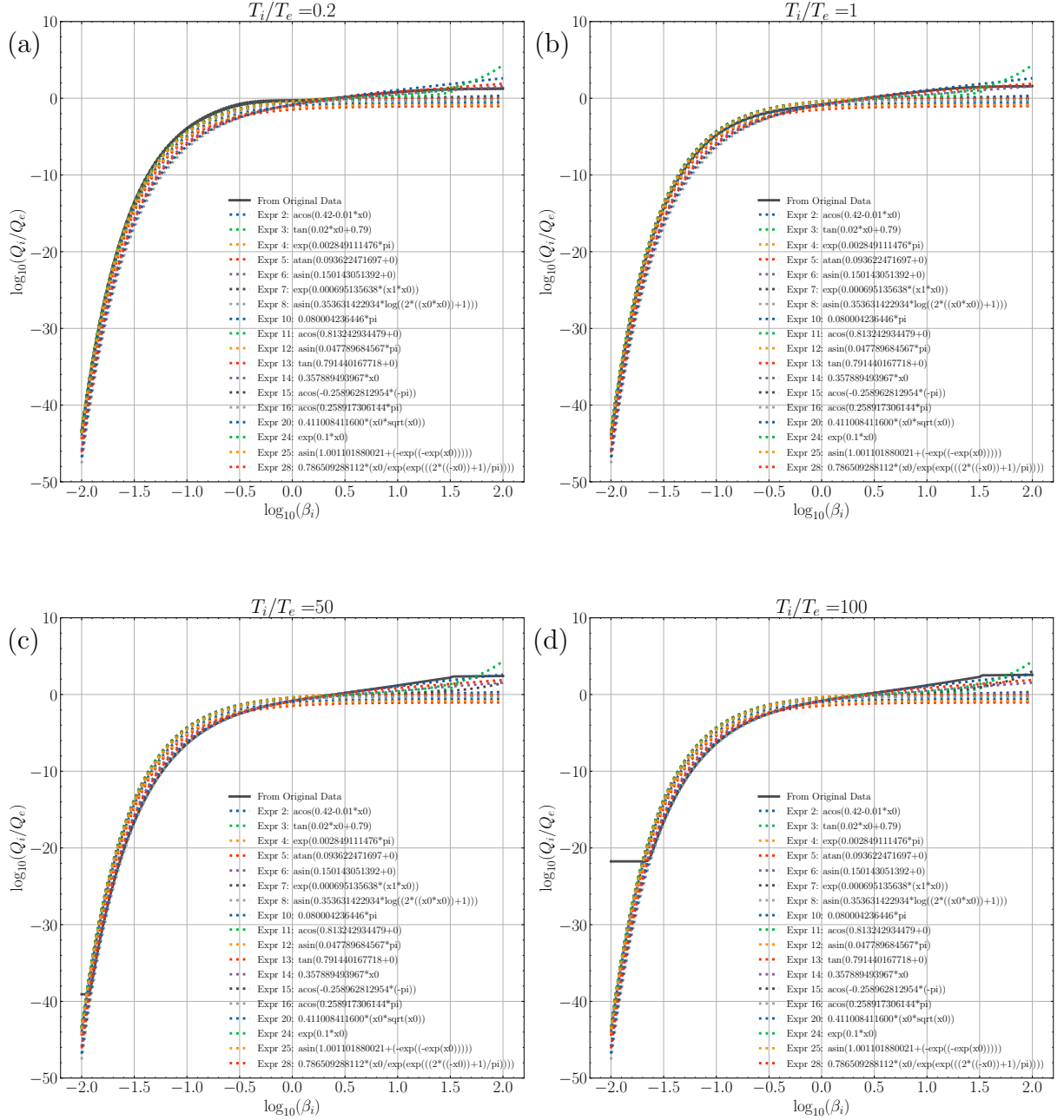


Figure 6: Selected symbolic expressions with the best performance from AI Feynman fitting with input data $\beta_i, T_i/T_e, e^{1/\beta_i}(Q_i/Q_e)$ are plotted with $T_i/T_e = 0.2, 1, 50, 100$. For all the expressions from this test, see Fig. 5 (d).

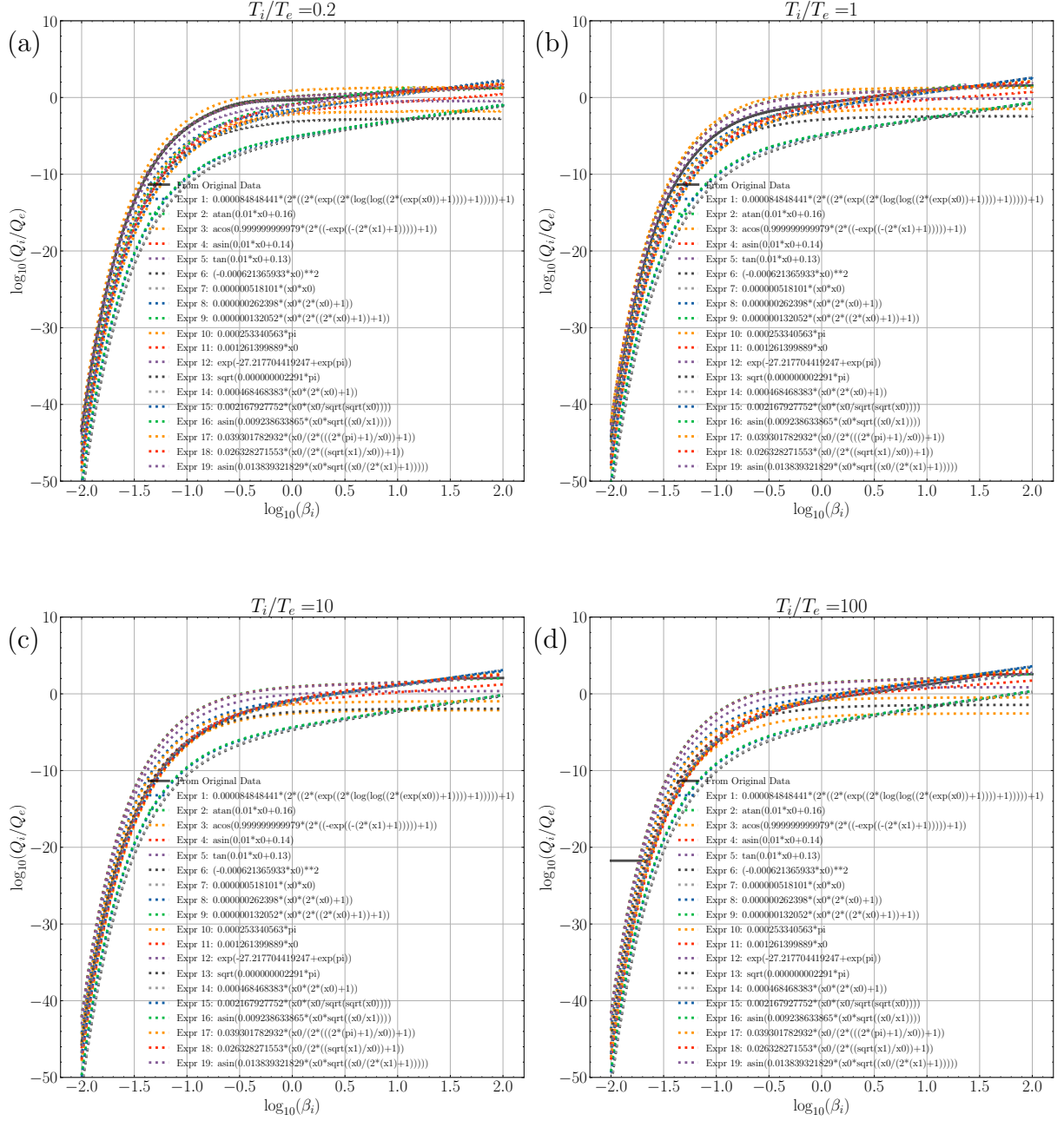


Figure 7: Results from AI Feynman fitting with input data $\beta_i, T_i/T_e, \sqrt{m_e T_e/(m_i T_i)} e^{1/\beta_i} (Q_i/Q_e)$, plotted with $T_i/T_e = 0.2, 1, 10, 100$.

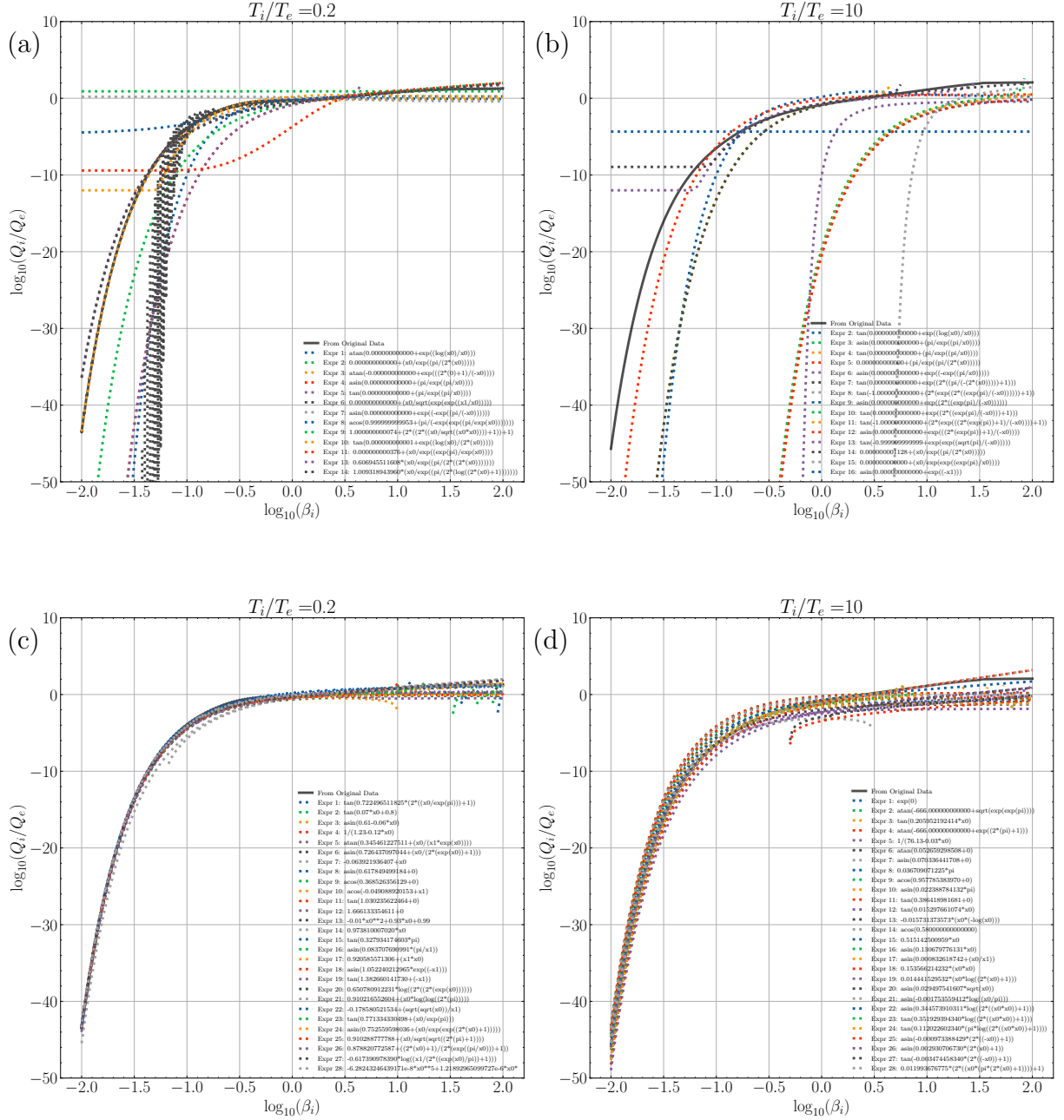


Figure 8: Results from AI Feynman fitting with input data segmented by the value of T_i/T_e . Left column: input data with $T_i/T_e < 1$; right column: input data with $T_i/T_e > 1$. Top row: input data $\beta_i, T_i/T_e, Q_i/Q_e$; bottom row: input data $\beta_i, T_i/T_e, e^{1/\beta_i} Q_i/Q_e$.

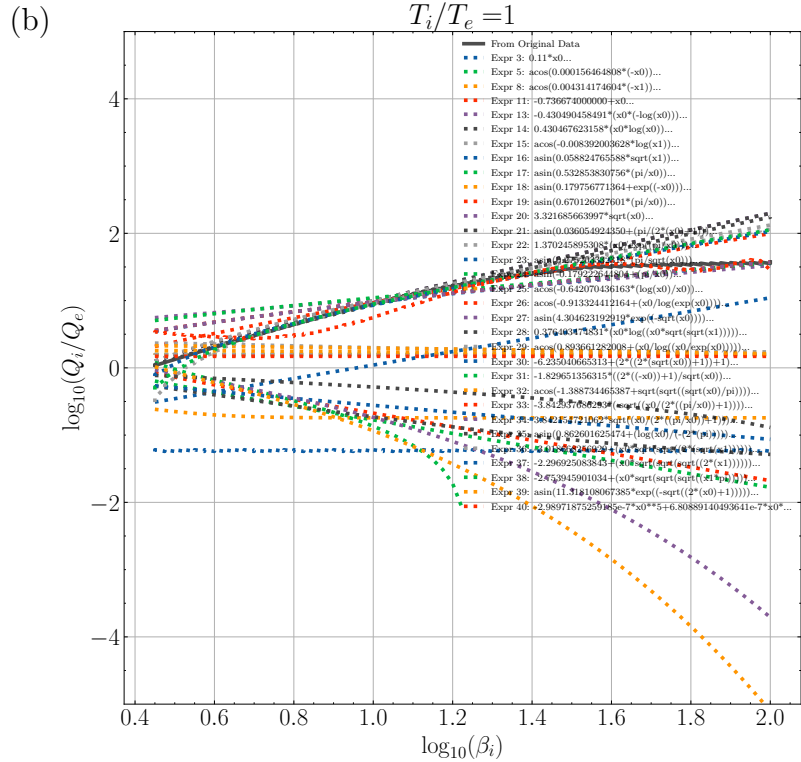
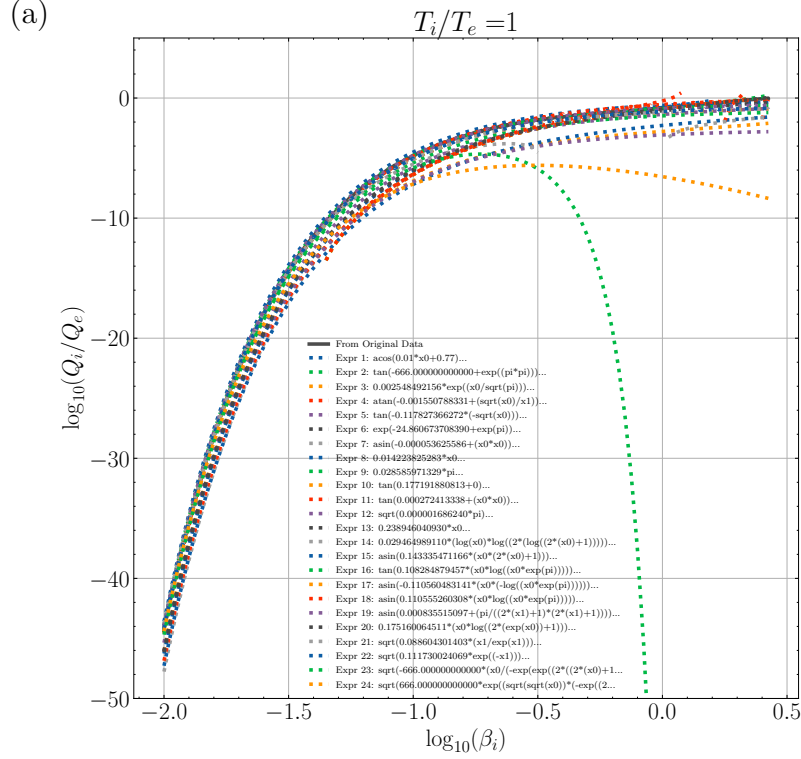


Figure 9: Results from AI Feynman fitting with input data segmented by the value of Q_i/Q_e . (a): input data $\beta_i, T_i/T_e, e^{1/\beta_i} Q_i/Q_e$ with $Q_i/Q_e < 1$; (b): input data with $\beta_i, T_i/T_e, Q_i/Q_e$ with $Q_i/Q_e > 1$.

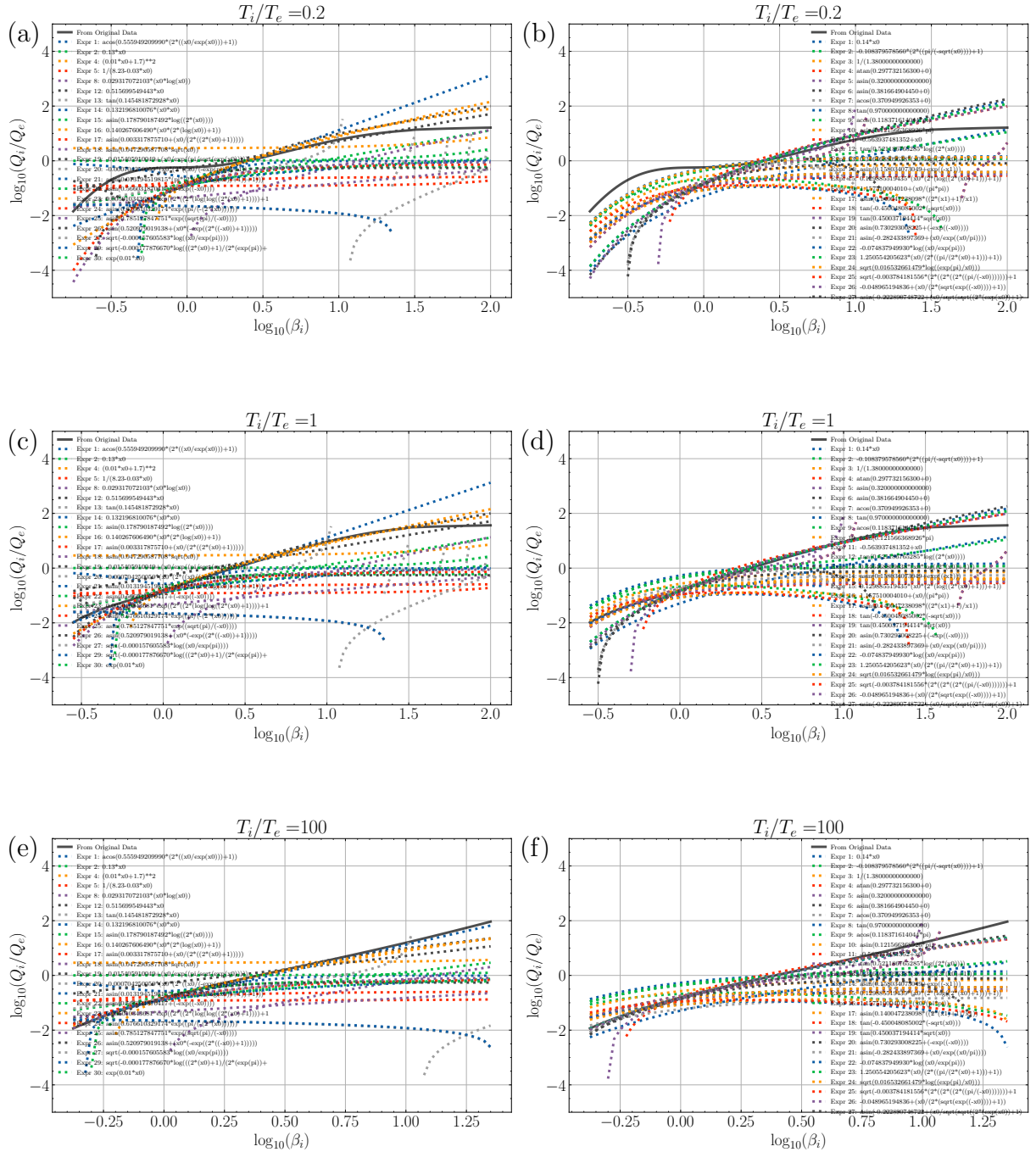


Figure 10: Results from AI Feynman fitting with input data where $0.01 < Q_i/Q_e < 100$. Left column: input data $\beta_i, T_i/T_e, Q_i/Q_e$; right column: input data $\beta_i, T_i/T_e, e^{1/\beta_i} Q_i/Q_e$

7 Discussion

Takeaways of this study:

1. **Overall Performance of AI Feynman:** To answer the question posed in §3: No, AI Feynman does not outperform the human-fitted empirical formula Eq.(1) at least in this course project. Specifically, AI Feynman struggles to capture the weak dependence on T_i/T_e . Most of the generated expressions do not incorporate T_i/T_e as a related variable. The test that results in the highest incidence of T_i/T_e involves passing $\beta_i, T_i/T_e, e^{1/\beta_i} Q_i/Q_e$ for $T_i/T_e < 1$, as illustrated in Fig. 8(c).
2. **Data Transformation Impact:** Using input data where the original dependent variable Q_i/Q_e is divided by the dominant term (i.e. passing $\beta_i, T_i/T_e, e^{1/\beta_i} Q_i/Q_e$ as the input data) consistently helps the AI Feynman capture the overall trend more effectively. This effect is evident in the comparison between Fig. 5 (d) and the other three panels of Fig. 5, as well as between the top and the bottom rows of Fig. 8.
3. **Concerns of the Description Length:** AI Feynman’s output is picked by minimizing the description length for both the error dataset (MEDL) and the expression (Complexity). As noted earlier in §5, the complexity of the empirical formula, according to their definition, is 95.77, which is very large compared to the average complexity of AI Feynman outputs. This may hinder AI Feynman from generating a similar formula due to its inherent penalty on complexity. Another concern is that a low MEDL does not necessarily correlate with better performance. In Fig. 11, I plotted the MEDL and complexity of all 14 expressions generated by AI Feynman in the test without data preprocessing (these 14 expressions themselves are plotted and printed in Fig. 4(a)). Although half of these expressions exhibit a lower MEDL than the empirical formula, none match its performance in fitting the original data.
4. **Limitations of this Project:** The scope of exploration in this project is quite restricted. First, the AI Feynman code is utilized as a “black box” without any deeper understanding or modification. Second, there is a lack of a quantifiable method to evaluate the performance of the expressions. The MEDL method provided by AI Feynman publications does not suffice as a standard measure. Third, considering that the big picture is to develop an enhanced plasma turbulent heating model, our goal is therefore not to improve a specific algorithm from a data science perspective, but to apply these algorithms to discover new physics insights into plasma turbulence heating. This suggests a need for further tests of various symbolic regression algorithms.

8 Some Personal Comments

In this project, AI Feynman did not outsmart the human scientist. Is this a deadend for the use of employing scientific machine learning methods in our research?

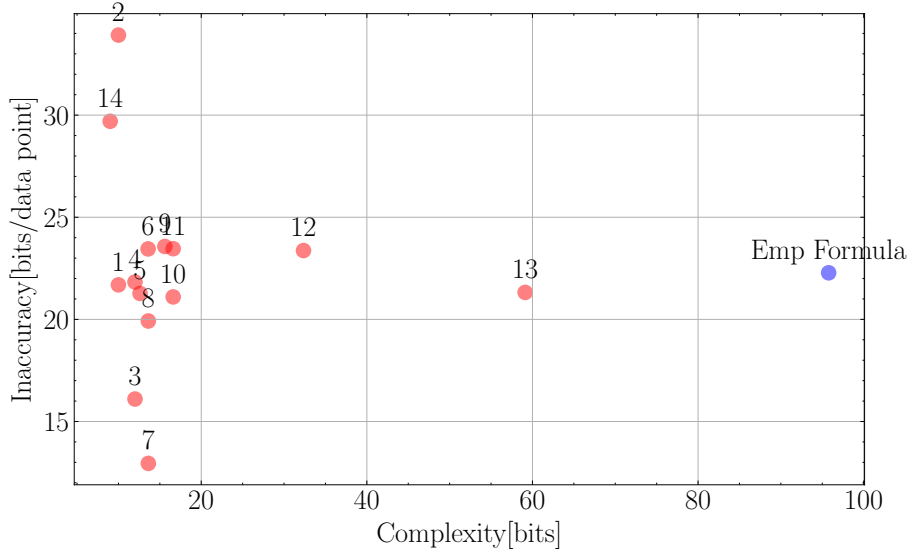


Figure 11: Inaccuracy (MEDL) versus Complexity Plot for the empirical formula Eq.(1) and 14 symbolic expressions from AI Feynman without data pre-processing, i.e. expressions in Fig. 4(a).

As the person who conducted this project and the author of this report, I believe that it is too early to draw any definitive conclusions about the future utility of scientific machine learning methods for our goal. First, as discussed earlier in §7, this project is significantly limited, and there are numerous possibilities for improvements. Second, even within this very immature initial work, the finding that the strategy of incorporating dominant terms—often derived from physical intuition—into the input data helps the algorithm capture the overall trends, may prove valuable in guiding our future research.

Third, I personally believe that much of human misery and suffering stems from two fundamental facts: (1) the limited lifespan of one human individual, and (2) the inability of individuals to interchange information at a cognitive level similar to data transmission between two computers, i.e. human cannot directly read each other’s mind. In the case of scientific discovery, the first limitation bounds the total possible scientific contributions that an individual can make over their lifetime. The second limitation affects how efficiently individuals can share and process information, directly impacting the rate at which a person can learn and, consequently, the pace at which science can advance. As the body of fundamental knowledge grows, the slow rate of information transmission between individuals becomes increasingly evident. In earlier times, when there was less accumulated knowledge, notable contributions to science could be made early in a person’s career—consider de Broglie, who won the Nobel Prize for his PhD thesis. Today, the typical outcome of completing a PhD is merely to obtain basic scientific knowledge with a small measure of novelty, with significant research contributions generally occurring much later. Combined with the depressing fact that the lifespan of a human individual is not only limited but also includes a significant period during the aging years when cognitive abilities notably decline, there is a pressing need to revolutionize the process of scientific discovery and even the structure of human society.

Meanwhile, recent years have witnessed the rapid rise of machine learning and AI technol-

ogy. From the perspective of humanity, the concept of artificial general intelligence (AGI), although not yet achieved, closely mirrors the cognitive abilities of humans (Tegmark, 2018) and may hold the key to effectively inheriting and advancing human scientific knowledge and even the entirety of civilization. From my own perspective for our research on plasma turbulent heating, for a long time, I have harbored the aspiration to use scientific machine learning not only to effectively store scientific discoveries but also to facilitate further advancements in science. Specifically, a recent study on pre-trained machine learning models for science (McCabe et al., 2023) has caught my attention. This research is inspired by the concepts of foundation models and transfer learning. A foundation model is a machine learning model trained on a broad range of data, enabling it to be applied widely. In simpler terms, the training method for a foundation model involves, for instance, teaching a model both English and French before fine-tuning it to specialize in translation, rather than training it solely on translation tasks. ChatGPT is a famous successful example of training foundation models. This is the motivation for the development of Polymathic AI, available at <https://polymathic-ai.org/>. Scientists are training this model on fundamental physics to ensure it doesn't start from scratch when tackling real-world problems. The findings are impressive: McCabe et al. (2023) demonstrates that even models pre-trained with non-scientific data like cat videos outperform those without any pre-training in solving scientific problems. The development of Polymathic AI is still ongoing, but it will always remain open-source, providing another option for our research.

I am fully aware that this last section reflects personal opinions, which may not hold true if more evidence is taken into consideration. Nonetheless, I have chosen to include them because they genuinely represent the motivation behind my passion for scientific machine learning.

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References

- Brunton, S. L., Proctor, J. L., and Kutz, J. N. (2016). Discovering governing equations from data by sparse identification of nonlinear dynamical systems. *Proceedings of the national academy of sciences*, 113(15):3932–3937.
- Cowan, N. (2010). The magical mystery four: How is working memory capacity limited, and why? *Current directions in psychological science*, 19(1):51–57.
- Cranmer, M., Sanchez Gonzalez, A., Battaglia, P., Xu, R., Cranmer, K., Spergel, D., and Ho, S. (2020). Discovering symbolic models from deep learning with inductive biases. *Advances in Neural Information Processing Systems*, 33:17429–17442.
- Einstein, A. (1916). The foundation of the general theory of relativity. *Annalen Phys*, 49(7):769–822.
- Grünwald, P. D., Myung, I. J., and Pitt, M. A. (2005). *Advances in minimum description length: Theory and applications*. MIT press.

- Hawking, S. (2002). *On the shoulders of giants: The great works of physics and astronomy*.
- Heeffer, A. (2011). Models of scientific discovery: what do they explain?
- Hernandez, A., Balasubramanian, A., Yuan, F., Mason, S. A., and Mueller, T. (2019). Fast, accurate, and transferable many-body interatomic potentials by symbolic regression. *npj Computational Materials*, 5(1):112.
- Howes, G. G. (2010). A prescription for the turbulent heating of astrophysical plasmas. *Monthly Notices of the Royal Astronomical Society: Letters*, 409(1):L104–L108.
- Howes, G. G. (2024). The fundamental parameters of astrophysical plasma turbulence and its dissipation: Nonrelativistic limit. *arXiv preprint arXiv:2402.12829*.
- Howes, G. G., Cowley, S. C., Dorland, W., Hammett, G. W., Quataert, E., and Schekochihin, A. A. (2008). A model of turbulence in magnetized plasmas: Implications for the dissipation range in the solar wind. *Journal of Geophysical Research: Space Physics*, 113(A5).
- Jin, Y., Fu, W., Kang, J., Guo, J., and Guo, J. (2019). Bayesian symbolic regression. *arXiv preprint arXiv:1910.08892*.
- Kolmogorov, A. N. (1963). On tables of random numbers. *Sankhyā Ser. A*, 25:369–376.
- Koza, J. R. (1994). Genetic programming as a means for programming computers by natural selection. *Statistics and computing*, 4:87–112.
- La Cava, W., Orzechowski, P., Burlacu, B., de Franca, F. O., Virgolin, M., Jin, Y., Komenda, M., and Moore, J. H. (2021). Contemporary symbolic regression methods and their relative performance. In *Thirty-fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track (Round 1)*.
- Lundberg, S. M. and Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in neural information processing systems*, 30.
- Majumder, M., Cranmer, M., Petersen, B., et al. (2023). Interpretable symbolic regression for data science: Analysis of the 2022 competition. *arXiv preprint arXiv:2304.01117*.
- McCabe, M., Blancard, B. R.-S., Parker, L. H., Ohana, R., Cranmer, M., Bietti, A., Eickenberg, M., Golkar, S., Krawezik, G., Lanusse, F., et al. (2023). Multiple physics pretraining for physical surrogate models. *arXiv preprint arXiv:2310.02994*.
- Miller, G. A. (1956). The magical number seven, plus or minus two: Some limits on our capacity for processing information. *Psychological review*, 63(2):81.
- Moseley, B. (2022). *Physics-informed machine learning: from concepts to real-world applications*. PhD thesis, University of Oxford.

- Petersen, B. K., Landajuela, M., Mundhenk, T. N., Santiago, C. P., Kim, S. K., and Kim, J. T. (2019). Deep symbolic regression: Recovering mathematical expressions from data via risk-seeking policy gradients. *arXiv preprint arXiv:1912.04871*.
- Quataert, E. (1998). Particle heating by alfvénic turbulence in hot accretion flows. *The Astrophysical Journal*, 500(2):978.
- Rudin, C. (2019). Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature machine intelligence*, 1(5):206–215.
- Schmidt, M. and Lipson, H. (2009). Distilling free-form natural laws from experimental data. *science*, 324(5923):81–85.
- Schmidt, M. D. and Lipson, H. (2011). Automated modeling of stochastic reactions with large measurement time-gaps. In *Proceedings of the 13th annual conference on Genetic and evolutionary computation*, pages 307–314.
- Strogatz, S. (2019). *Infinite powers: How calculus reveals the secrets of the universe*. Eamon Dolan Books.
- Tegmark, M. (2008). The mathematical universe. *Foundations of physics*, 38(2):101–150.
- Tegmark, M. (2018). *Life 3.0: Being human in the age of artificial intelligence*. Vintage.
- Udrescu, S.-M., Tan, A., Feng, J., Neto, O., Wu, T., and Tegmark, M. (2020). Ai feynman 2.0: Pareto-optimal symbolic regression exploiting graph modularity. *Advances in Neural Information Processing Systems*, 33:4860–4871.
- Udrescu, S.-M. and Tegmark, M. (2020). Ai feynman: A physics-inspired method for symbolic regression. *Science Advances*, 6(16):eaay2631.
- Virgolin, M. and Pissis, S. P. (2022). Symbolic regression is np-hard. *Transactions on Machine Learning Research*.
- Wu, T. and Tegmark, M. (2019). Toward an artificial intelligence physicist for unsupervised learning. *Physical Review E*, 100(3):033311.